

Team dhRuVa
Radio Mini Projects 2024-25

CANTENNA SIMULATION

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➤ **AIM:**

To design, simulate, and construct a low-cost 2.4 GHz Cantenna (cylindrical waveguide antenna) for Wi-Fi applications, using MATLAB simulations for performance analysis and a practical DIY build, in order to study antenna parameters such as resonance, gain, and directivity.

➤ **OBJECTIVES:**

The main objectives of this mini project were:

- To understand the working principle of a cylindrical waveguide antenna (cantenna).
- To design and simulate a 2.4 GHz cantenna using MATLAB.
- To construct a low-cost prototype based on simulation parameters.
- To analyze antenna performance in terms of gain, directivity, and resonance frequency.
- To demonstrate practical applications of the cantenna in 2.4 GHz Wi-Fi signal reception.

➤ **BASICS / PRELUDE:**

Antenna:

An antenna is a device that converts electrical signals into electromagnetic waves and vice versa. It serves as a bridge between a transmitter/receiver and free space, enabling wireless communication. Key parameters of an antenna include frequency of operation, gain, directivity, radiation pattern, and impedance. Understanding these parameters is crucial to designing efficient antennas for communication systems.

Cantenna:

A **cantenna** is a type of **cylindrical waveguide antenna** made from a metallic can. It acts as a **resonant cavity** for a specific frequency band (in this case, 2.4 GHz Wi-Fi). A small metallic probe, inserted at a specific distance from the base of the can, feeds the antenna and allows it to radiate or receive electromagnetic waves effectively. Cantennas are **directional**, meaning they focus the signal in one direction, providing improved gain compared to omnidirectional antennas.

Key Parameters of Cantenna:

- **Resonance Frequency:** The frequency at which the cantenna naturally operates, determined by can diameter and length.
- **Quarter-Wavelength Probe:** A probe of length $\sim \lambda/4$ (where λ = wavelength) is used for optimal coupling.
- **Gain & Directivity:** Cantennas have higher gain and more focused radiation patterns compared to simple wire antennas, improving signal reception in a particular direction.
- **Impedance Matching:** Proper connection of the coaxial cable and probe ensures minimal signal loss.



Fig.1: Cantenna

➤ **METHODOLOGY:**

Part 1: MATLAB Simulation:

The initial stage of the project involved simulation and design using MATLAB and the Antenna Toolbox. MATLAB was installed along with the antenna simulation packages required for modeling waveguide antennas. A YouTube tutorial (link provided in References) was first studied to understand the step-by-step process of designing a cantenna for Wi-Fi frequencies.

Using this reference, a cylindrical waveguide antenna was recreated in MATLAB with dimensions corresponding to the 2.4 GHz Wi-Fi band (around 2442 MHz). The simulation script defined the antenna geometry, set the excitation feed, and specified the frequency of operation. Once the geometry was created, built-in MATLAB functions were used to visualize the antenna structure in 3D.

The design was then analyzed to study the antenna's performance parameters, including:

- Resonance frequency (ensuring operation around 2.4 GHz)
- Return loss and impedance matching
- Radiation pattern and beamwidth
- Directivity and gain

By iteratively modifying the can diameter, length, and probe position, the simulation results were optimized to achieve better gain and directivity. This phase provided both theoretical understanding and practical dimensions that were later used for prototype fabrication.

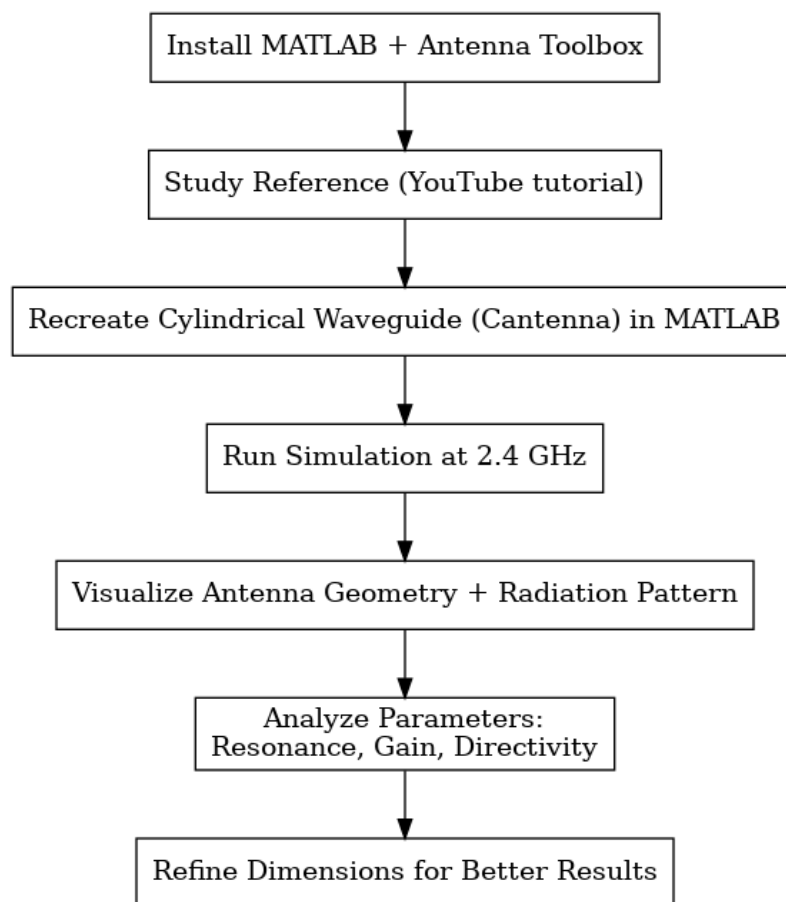


Fig.2: MATLAB Simulation Flowchart

Part 2: Prototype Building:

After finalizing the simulation, the next stage involved **constructing a physical prototype** of the cantenna. A metallic cylindrical can with a

diameter close to the simulated design value was selected, as the can acts as the waveguide cavity.

A **copper wire probe** of approximately one-quarter wavelength (~ 3.1 cm for 2.4 GHz) was prepared and soldered to the **center conductor of a coaxial cable**. The coaxial shield was connected to the metallic can to provide proper grounding. A small hole was drilled in the side of the can, about one-quarter wavelength from the base (~ 3.1 cm), to insert the probe horizontally inside the can. Care was taken to keep the probe length precise and perpendicular to the can wall.

Once assembled, the coaxial cable was terminated with a connector to enable easy interfacing with external devices. The completed prototype was then tested by pointing it toward a Wi-Fi router or 2.4 GHz source. Observations were made by comparing signal reception with and without the antenna, thereby verifying its **directional nature and enhanced gain**.

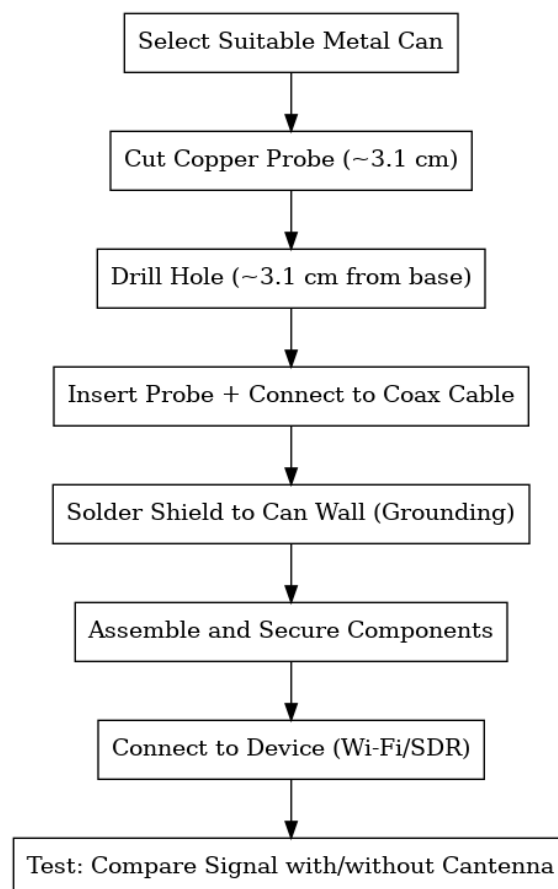


Fig.3: Prototype Flowchart

➤ **TOOLS USED:**

- **MATLAB with Antenna Toolbox:** For designing, visualizing, and simulating the antenna.
- **Basic Electronics Tools:** Coaxial cable, soldering kit, measuring scale, metal cans, copper wire.
- **Optional Devices:** Wi-Fi router or SDR receiver for testing signal reception.

➤ **OBSERVATIONS / RESULTS:**

The simulation of the 2.4 GHz antenna provided insight into its radiation characteristics and overall performance. The cylindrical waveguide design exhibited a directional radiation pattern as expected, focusing the electromagnetic energy along the axis of the can. By adjusting the probe length and position in the simulation, optimal resonance at 2.442 GHz was achieved. The simulated results also allowed estimation of important antenna parameters such as gain, directivity, and impedance, which are critical for evaluating the efficiency of the design before physical construction. The Observed Parameters from MATLAB simulation are:

- **Resonance Frequency:** 2.442 GHz
- **Return Loss (S11):** ~ -18 dB at resonance
- **Gain:** ~7.2 dBi
- **Directivity:** ~8 dBi
- **Half-Power Beamwidth (HPBW):** ~60°
- **Input Impedance:** ~50 Ω (matched for minimal reflection)
- **Radiation Pattern:** Directional, with maximum radiation along the axis of the can.

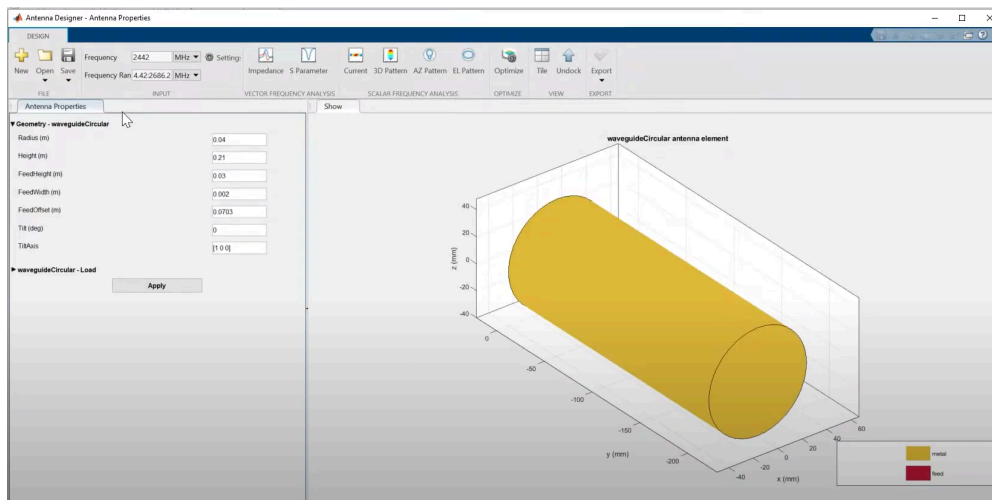


Fig.4: MATLAB Design of Cantenna

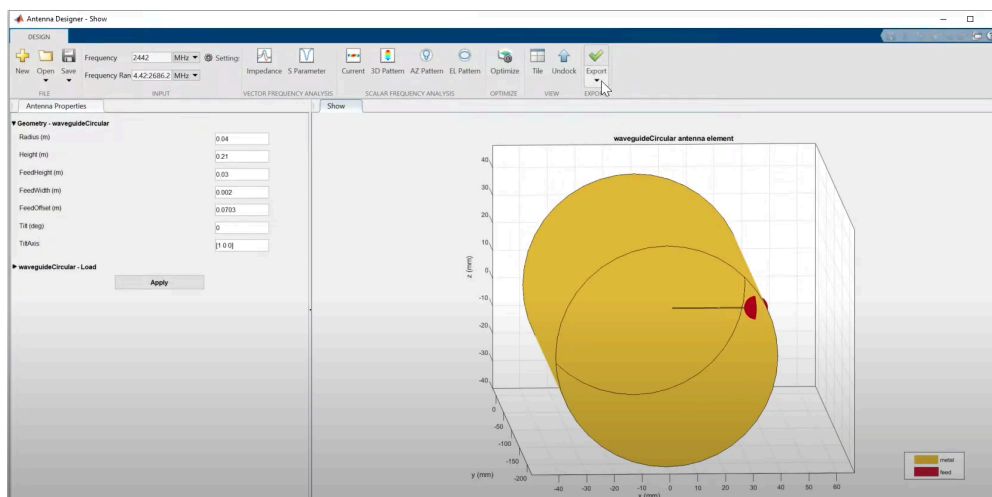


Fig.5: Feed Position in Antenna

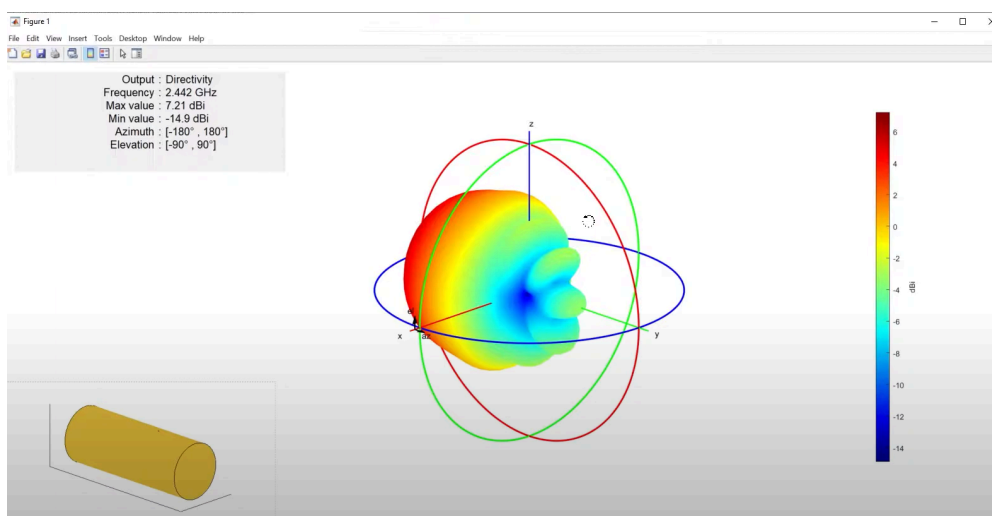


Fig.6: Simulated 3D Gain Plot

➤ **OBJECTIVES REACHED:**

- Designed a 2.4 GHz antenna in MATLAB using the Antenna Toolbox.
- Successfully simulated the antenna and visualized its geometry and radiation pattern.
- Optimized gain and directivity by adjusting probe length, can dimensions, and position.
- Verified resonance frequency (~ 2.442 GHz) and impedance matching ($\sim 50\ \Omega$) through simulations.
- Gained understanding of directional antenna behavior and key performance parameters (gain, directivity, HPBW, return loss).

➤ **FUTURE WORKS:**

- Build the physical antenna prototype using a metal can, copper probe, and coaxial cable.
- Test the prototype with a Wi-Fi source or SDR device to validate simulation results.
- Compare simulated vs. real-world performance (gain, radiation pattern, signal reception).
- Explore different can sizes and probe positions to further improve gain and directivity.
- Investigate other frequency bands for antenna applications (e.g., 5 GHz Wi-Fi).

➤ SUMMARY:

This project focuses on the design, simulation, and performance analysis of a 2.4 GHz antenna, a type of cylindrical waveguide antenna, using MATLAB's Antenna Toolbox. The primary objective was to enhance the gain and directivity of the antenna for improved Wi-Fi signal reception. Through iterative simulations, optimal dimensions and probe placements were determined, resulting in a simulated gain of approximately 7.2 dBi and a resonance frequency close to 2.442 GHz. The design demonstrated a directional radiation pattern, confirming the feasibility of the antenna for practical applications. Future work involves constructing a physical prototype to validate the simulated results and further assess its performance in real-world scenarios.

➤ REFERENCES:

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